Algebra 2 *Advanced* Unit 6 Bellwork: Week 1

Name: ____________________ Date: __________ Period: ______

Day 1: Prior Knowledge Review

1. Simplify completely: a) $\sqrt[3]{-125} =$ ☐ b) $\sqrt[3]{-\frac{8}{27}} =$ ☐	**2.** Rewrite using fractional exponents: a) $\sqrt[3]{x^6} =$ ☐ b) $\left(8x^3\right)^{2/3} =$ ☐
3. [SAT] Line ℓ passes through $(3, -2)$ and is perpendicular to $2x - 5y = 10$. What is the y-intercept of line ℓ? A) $-\frac{19}{2}$ B) $\frac{11}{2}$ C) $\frac{7}{2}$ D) $\frac{19}{2}$ Answer: ☐	**4. [SAT]** If $\frac{3x-2}{4} - \frac{x+1}{3} = 2$, what is x? $x =$ ☐

Day 2: Review - Basic Cube Root Solving

1. Solve: $\sqrt[3]{2x-5} = -3$ $x =$ ☐	**2.** Solve: $\sqrt[3]{x^2-9} = \sqrt[3]{2x+6}$ $x =$ ☐
3. Solve: $\sqrt[3]{4x+1} - \sqrt[3]{2x-3} = 0$ $x =$ ☐	**4. [SAT]** The line $y = kx + 4$ passes through (a, b) where $a \neq 0$. The line $y = \frac{x}{k} + 4$ also passes through (a, b). Which must be true? A) $k = 1$ or $k = -1$ B) $k = 0$ C) $a = b$ D) $b = 4$ Answer: ☐

Day 3: Review - Multi-Step & Exponents

1. Solve and CHECK: $(2x+1)^{1/3} - 3 = 0$

$x =$ ☐

Check:

2. Solve: $\frac{1}{2}\sqrt[3]{3x-4} + 5 = 3$

$x =$ ☐

3. Solve for x in terms of a and b:
$(x-a)^{1/3} = b$

$x =$ ☐

4. **[SAT]** If $-3 < \frac{2x-1}{3} < 5$, which represents all values of x?

A) $-4 < x < 8$ B) $-5 < x < 7$

C) $-4 < x < 6$ D) $-\frac{5}{2} < x < \frac{13}{2}$

Answer: ☐